

Johnson-Nyquist Noise: A Tutorial

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This notebook accompanies two papers published back to back in *Physical Review* **32** (July 1928):

- J. B. Johnson, “Thermal Agitation of Electricity in Conductors,” *Phys. Rev.* **32**, 97 (1928) – the experimental discovery.
- H. Nyquist, “Thermal Agitation of Electric Charge in Conductors,” *Phys. Rev.* **32**, 110 (1928) – the thermodynamic derivation.

Johnson, at Bell Labs, had flagged the effect in two 1927 notes as a resistance-dependent component of amplifier “noise” distinguishable from tube shot noise. Nyquist’s paper, submitted after seeing Johnson’s results, derives the noise spectrum from the second law of thermodynamics

and equipartition alone, with no assumption about the microscopic nature of the charge carriers.

The target result, reached from two directions in this notebook:

$$\overline{E_\nu^2} = 4Rk_B T$$

the mean-square thermal EMF spectral density (units V^2/Hz) of a resistor R at temperature T – flat in frequency (white noise) in this classical limit. The mean-square voltage in a band $d\nu$ is then $\overline{E_\nu^2} d\nu$; the frequency dependence shows up only once R itself varies with ν (Section 6) or once the classical $k_B T$ -per-mode is replaced by the Planck oscillator average (Section 7).

Notebook structure

1. Physical background: equipartition and the fluctuation-dissipation theorem
2. Reproducing Johnson’s comparative measurements (synthetic data)
3. The RC-shunt rolloff
4. Boltzmann-constant extraction from a model amplifier
5. Nyquist’s thermodynamic argument (transmission line, mode counting)
6. Detailed balance / the second-law argument
7. Generalization to arbitrary networks
8. Classical-to-quantum crossover
9. Synthesis and further reading

1 Physical background: equipartition and the fluctuation-dissipation theorem

Two ideas from statistical mechanics do essentially all of the physical work in both papers, and it is worth having them stated plainly before either paper’s own argument is walked through, since neither is likely to be familiar coming from an imaging-science rather than a statistical-mechanics background.

The equipartition theorem. A system in thermal equilibrium at temperature T distributes its energy among its available *quadratic* degrees of freedom – any independent way the system can store energy that enters the total energy as the square of some variable (a velocity component, a displacement, a current, a charge) – and assigns each one, on average, exactly $\frac{1}{2}k_B T$ of energy, regardless of what that degree of freedom physically is. A gas molecule free to move in three dimensions has three such terms (one per velocity component) and so carries $\frac{3}{2}k_B T$ on average. A simple harmonic oscillator has two – one kinetic, one potential – and so carries a full $k_B T$ on average, split evenly between them. This second case is the one that matters here: each standing-wave mode on Nyquist’s transmission line (Section 5) behaves exactly like a harmonic oscillator, with its energy split between the electric field (playing the

role of potential energy) and the magnetic field (playing the role of kinetic energy) – $\frac{1}{2}k_B T$ each, $k_B T$ total per mode. The theorem requires no assumption about what is actually oscillating – electrons, ions, an electromagnetic field – which is precisely why Nyquist’s derivation needs no model of the charge carriers at all.

The fluctuation-dissipation theorem. This is the more general statement that a system’s capacity to *dissipate* energy into a thermal bath and its tendency to be randomly *driven* by that same bath are not two separate facts – they are the same fact, seen from two sides, and the theorem fixes the size of the random fluctuation directly in terms of the strength of the dissipation and the temperature. A resistor that can turn electrical current into heat (dissipation, quantified by R) is, by the same physical coupling, randomly agitated by the thermal motion of the charges and lattice around it, producing a fluctuating voltage across its terminals even when no current is deliberately driven through it at all (fluctuation) – and the theorem says the size of that fluctuation is fixed the moment R and T are fixed, with nothing else to adjust. The result this notebook derives twice, $\overline{E_\nu^2} = 4Rk_B T$, is a fluctuation-dissipation relation: R is the dissipative response, and $\overline{E_\nu^2}$ is the resulting equilibrium fluctuation spectrum it necessarily comes paired with. Johnson and Nyquist’s 1928 papers are, historically, the origin of this specific electrical instance of a principle that turns out to be completely general – the same reasoning fixes the size of Brownian motion in terms of a particle’s friction coefficient, and, closer to home for imaging science, fixes the noise floor of any real detector or amplifier stage in terms of its own dissipative loss mechanisms, which is why fluctuation-dissipation reasoning underlies noise-equivalent-power and detectivity calculations generally.

Both ideas get invoked at specific, marked points below: equipartition supplies the $k_B T$ per mode in Nyquist’s mode-counting argument (Section 5), and Section 6’s detailed-balance argument is what justifies treating that result as holding separately in *every* frequency band, rather than only on average – which is what makes $\overline{E_\nu^2}$ meaningful as a spectral density at all. Equipartition plus detailed balance together are exactly the two ingredients a fluctuation-dissipation relation requires, and Section 7 shows the result generalizing to any dissipative network, not just a bare resistor.

2 Setup

Set up the Jupyter notebook. This section will not appear in PDF document.

3 Reproducing Johnson’s comparative measurements

Johnson’s central empirical claims (his Figs. 4-6): the mean-square voltage fluctuation $\overline{V^2}$ across a resistor is proportional to R and to T , and is otherwise independent of the conductor’s material, shape, or conduction mechanism (metallic, electrolytic, light or heavy carriers).

Here we simulate his measurements by generating synthetic thermal-noise voltage traces directly from the target law and recover the scaling by simulated “measurement” – the same comparative-measurement logic Johnson used. The generator below targets a one-sided power spectral density of $4k_B T R$ (V^2/Hz); the total variance of a trace band-limited to `bandwidth_hz` is that density times the bandwidth, by Parseval’s theorem, which is what the rescaling step in the code enforces.

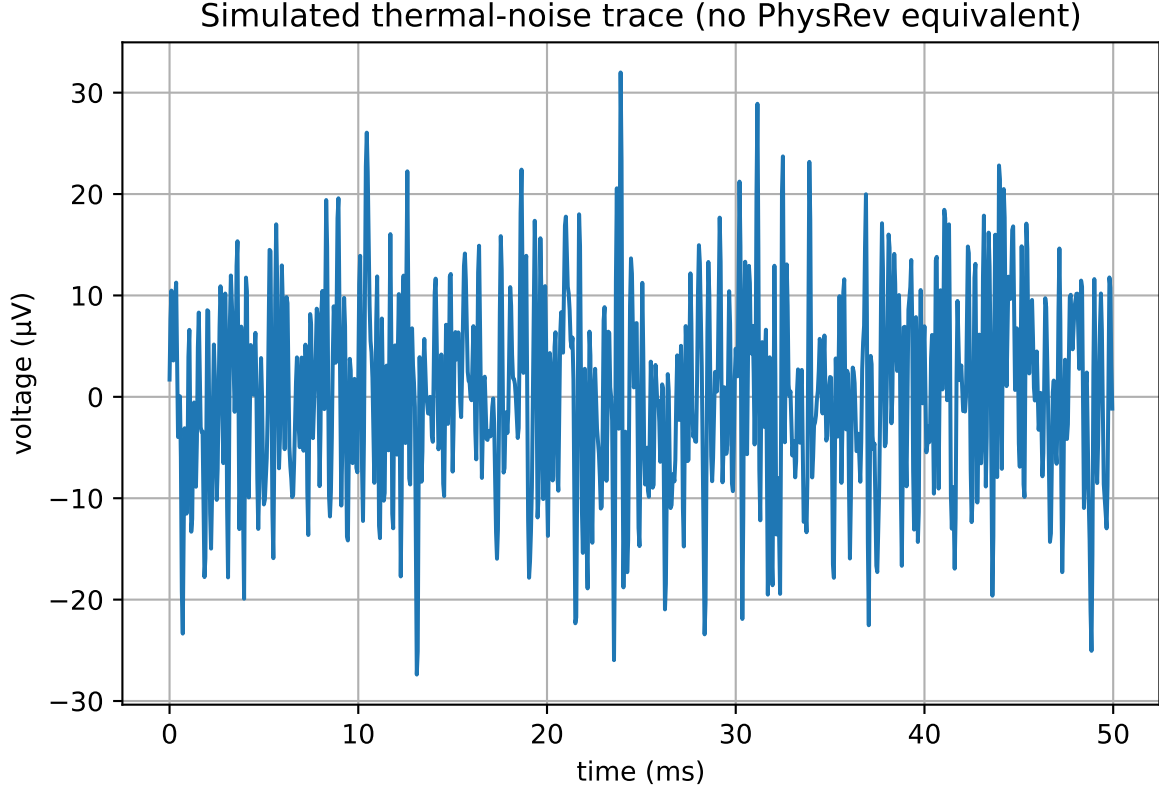


Figure 1: Simulated thermal-noise trace: $R = 1 \text{ M}\Omega$, $T = 300 \text{ K}$, bandwidth = 5 kHz (no corresponding PhysRev figure – illustrative only)

Johnson’s Fig. 4 held temperature and bandwidth fixed and varied the resistor’s material and construction (carbon filament, Advance wire, and several electrolytes) at each of several resistance values, showing that $\overline{V^2}$ collapses onto a single straight line in R regardless of conduction mechanism. Because the simulated generator above already encodes the noise law directly, “different conductors, same R ” reduces here to repeated draws at the same R ; the point under test is the recovered slope, which should equal $4k_B T \times \text{bandwidth}$.

```
fitted slope           = 8.284e-17 V^2/Ω
predicted 4 k_B T BW   = 8.284e-17 V^2/Ω
```

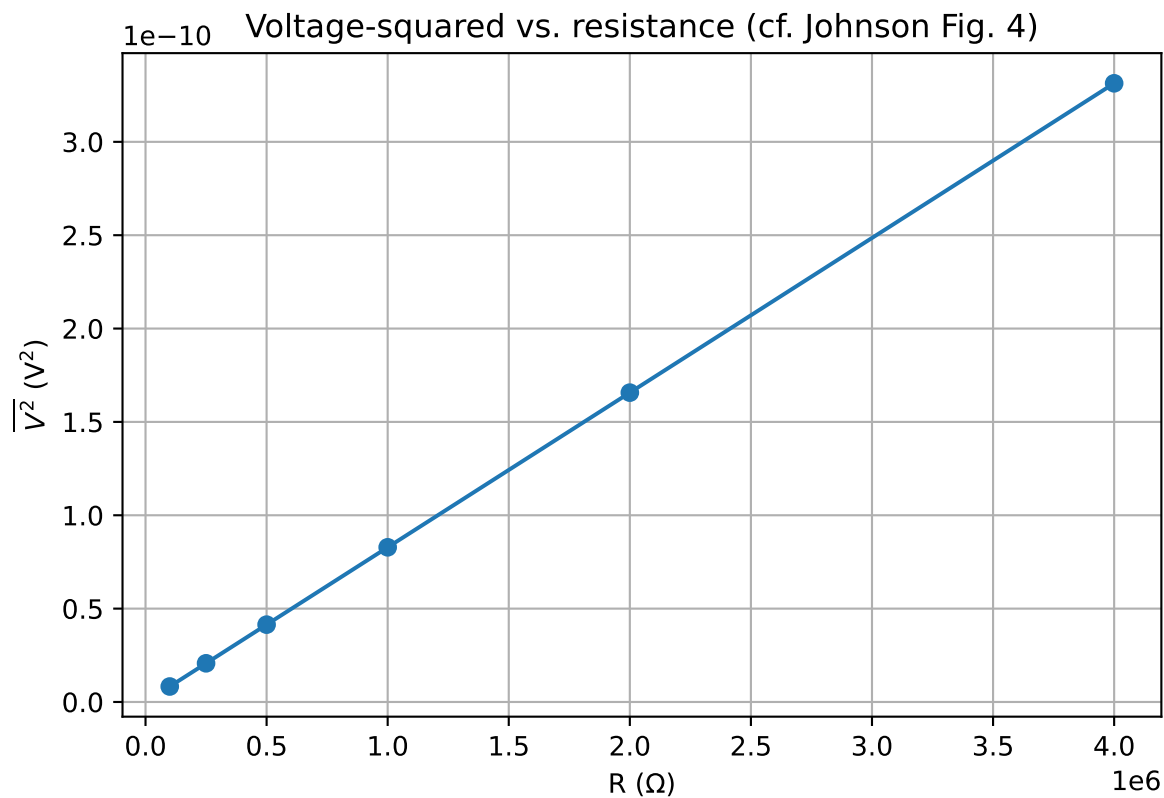


Figure 2: Voltage-squared vs. resistance (cf. Johnson, Phys. Rev. 32, 97, Fig. 4)

Johnson's Fig. 6 fixed the resistor and varied temperature instead, plotting the virtual power $W = \overline{V^2}/R$ – with R divided out, every conductor and every temperature should fall on one line through the origin with slope $4k_B \times \text{bandwidth}$.

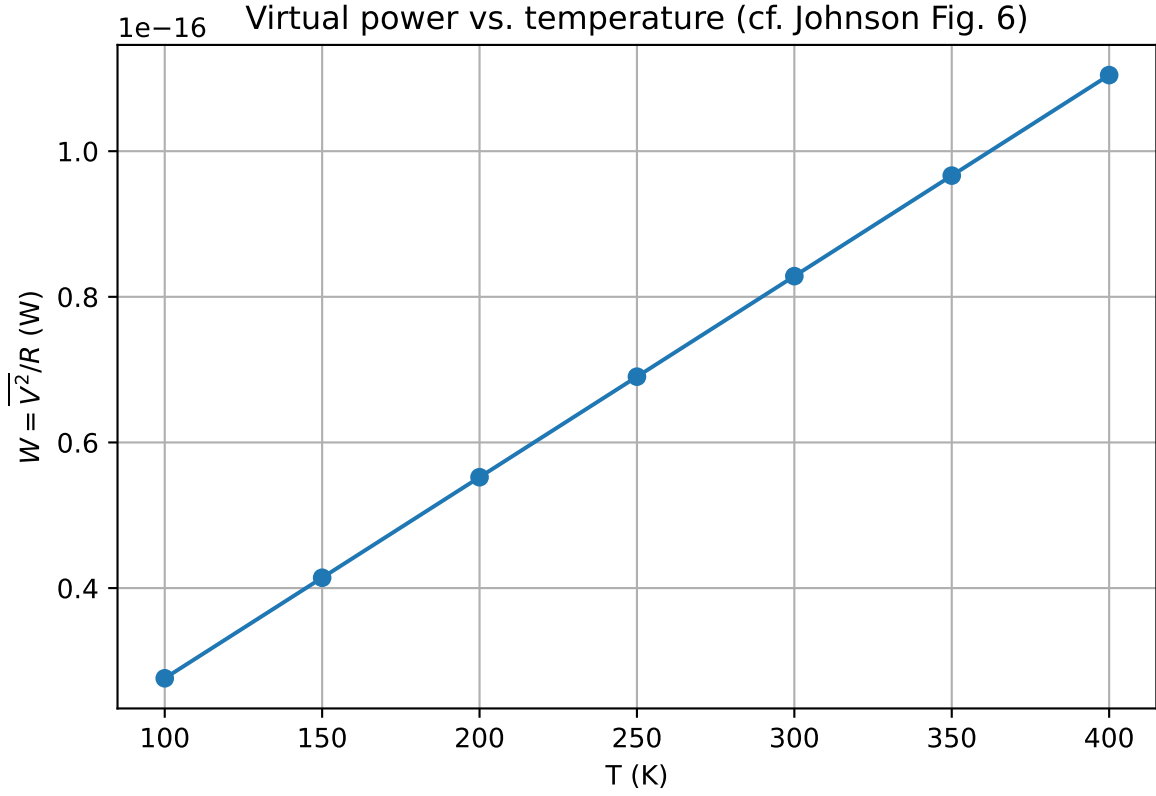


Figure 3: Virtual power vs. temperature (cf. Johnson, Phys. Rev. 32, 97, Fig. 6)

fitted slope = 2.761e-19 W/K
predicted 4 k_B BW = 2.761e-19 W/K

4 The RC-shunt rolloff

Johnson's apparatus had a real input resistance shunted by stray capacitance. With capacitance C across R_0 , the *real part* of the impedance is

$$R(\omega) = \frac{R_0}{1 + \omega^2 C^2 R_0^2}$$

so the apparent $\overline{V^2}$ rises with R_0 at fixed bandwidth only up to a point, then falls as R_0 is increased further – his Fig. 5.

This capacitance is a measurement artifact, not part of the physics being measured: it is the unavoidable stray capacitance of the resistor’s own leads plus the input (grid-to-filament) capacitance of the first amplifier tube, not something Johnson introduced deliberately. Its effect is to act as a low-pass filter in series with the true thermal-noise source, so at large R_0 the apparatus itself suppresses the measured signal even though the underlying noise law says $\overline{V^2}$ should keep growing with R without bound. Left unrecognized, the resulting turnover in Fig. 5 could easily be mistaken for a breakdown of the R -proportionality itself, rather than an artifact of the measuring circuit. Johnson instead measured R_0 and C directly for each resistance unit and showed that once this filtering is accounted for, the underlying law $\overline{V^2} \propto R$ continues to hold undisturbed – the calculation below reproduces that same accounting.

The parameters used (577 pF, 635 p.p.s.) are two of Johnson’s own quoted values for this measurement. The vertical axis is proportional to Boltzmann’s constant k_B (through the omitted bandwidth and temperature factors), so it is left in arbitrary units here; only the shape of the curve is being reproduced.

5 Boltzmann-constant extraction from a model amplifier

Johnson’s Eq. (1),

$$\overline{I^2} = \frac{2k_B T}{\pi} \int_0^\infty R(\omega) |Y(\omega)|^2 d\omega,$$

is how he actually extracted k_B (his Table I): measure $\overline{I^2}$, know T and $R(\omega)$, numerically integrate the amplifier’s measured $|Y(\omega)|^2$ characteristic, and solve for k_B .

Johnson needed his amplifier to respond strongly only within a narrow, adjustable band of frequencies, so that a single measured current could be related to the integral of $R(\omega)|Y(\omega)|^2$ over just that band – this is what one of his amplifier stages, a tunable resonant (LC) circuit, did in hardware: like a swing that only builds up a large amplitude when pushed near its natural frequency, an LC circuit passes signal efficiently only near its own resonant frequency and increasingly rejects frequencies farther from it. The “quality factor” Q is just a dimensionless number describing how narrow and sharply peaked that passband is: high Q means a narrow, sharply tuned response (very selective, but also slow to respond); low Q means a broad, loosely tuned one. Johnson’s real hardware response is his measured Fig. 3, reproduced by his instrumentation. Rather than model his actual vacuum-tube circuit in detail, we approximate that peaked, tunable response with a **Lorentzian** – a standard bell-shaped curve (the same shape that describes spectral line profiles in optics and resonance peaks generally) that closely matches the shape of a simple resonant circuit’s response near its peak, parameterized by just

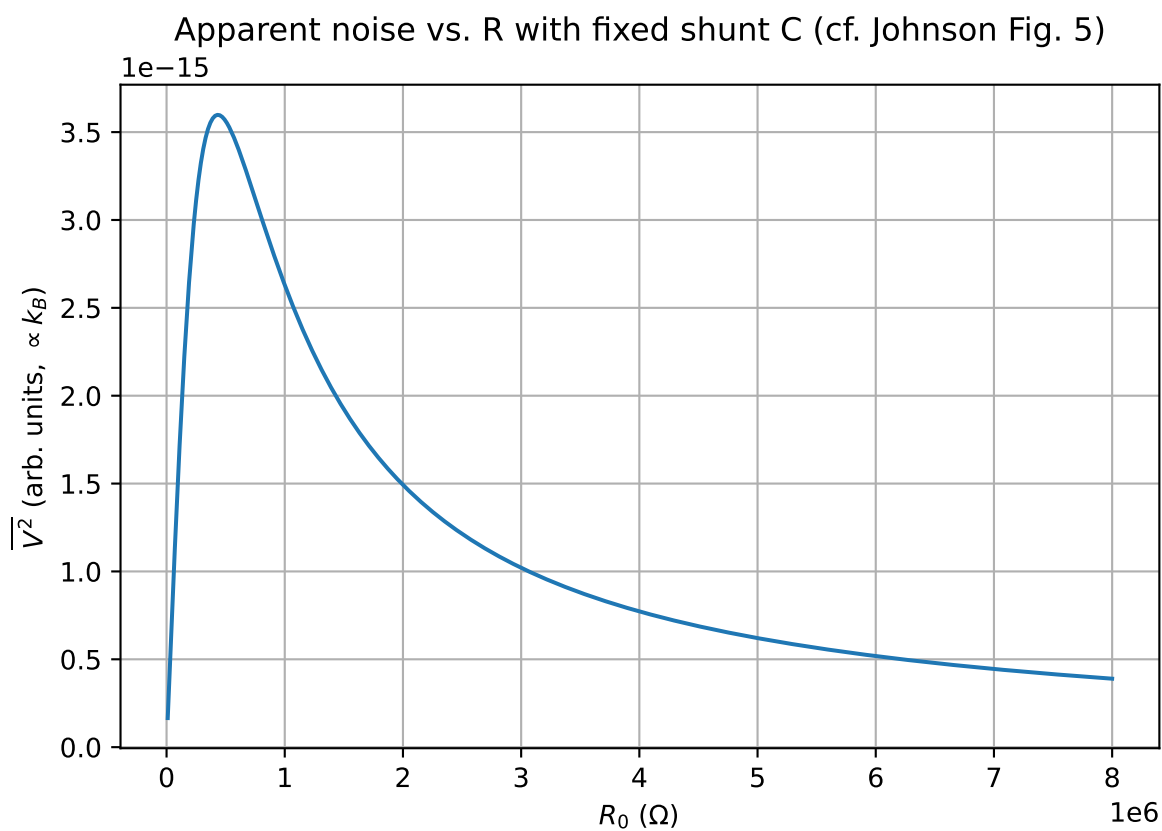


Figure 4: Apparent noise vs. R with fixed shunt C (cf. Johnson, Phys. Rev. 32, 97, Fig. 5)

a center frequency and Q . It is a convenient stand-in, not a claim about Johnson’s specific circuit components.

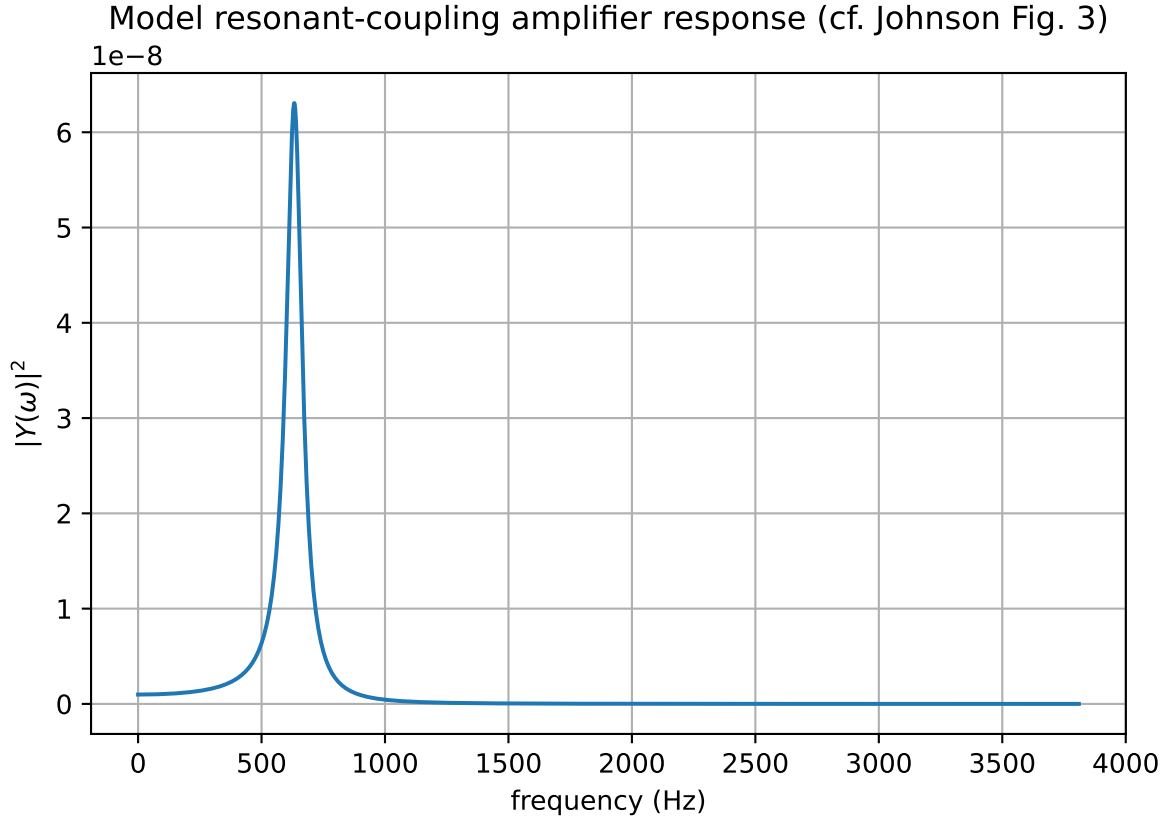


Figure 5: Model resonant-coupling amplifier response (cf. Johnson, Phys. Rev. 32, 97, Fig. 3)

$$|Y(\omega)|^2 \text{ d (numerical, truncated at } 6 \cdot 0) = 4.9206\text{e-}05$$

$R_0 = 5 \times 10^5 \Omega$ is comparable to Johnson’s “grid leak” input elements, of order one-half megohm. With the integral in hand, a synthetic “measured” $\overline{I^2}$ is generated from the *true* k_B plus multiplicative noise set to Johnson’s reported $\sim 13\%$ mean deviation, and Eq. (1) is inverted for k_B – reproducing one row of his Table I.

$$\begin{aligned} \text{true } k_B &= 1.3806\text{e-}23 \text{ J/K} \\ \text{recovered } k_B &= 1.6216\text{e-}23 \text{ J/K} \quad (+17.5\%) \end{aligned}$$

Repeating this across many synthetic runs, with f_0 , Q , R_0 , and T drawn randomly to stand in for Johnson’s 23 distinct measurement conditions, builds a table directly comparable to

his Table I, which listed 23 separate determinations of k_B made under different experimental settings.

Each row below is one such run: `f0_Hz` is the tuned resonant frequency of that measurement’s amplifier stage (Johnson varied this from about 300 to 2000 cycles per second across his experiments); `T_K` is the sample temperature for that run, near room temperature throughout; `R0_ohm` is the input resistor value used, standing in for Johnson swapping in different physical resistors between runs; and `k_B_fit` is the Boltzmann constant recovered by inverting Eq. (1) using that run’s own simulated $\bar{I}^2$, T , R_0 , and integrated amplifier response. `pct_error` is how far that single run’s estimate landed from the true constant.

No single row is expected to be accurate – each carries roughly Johnson’s own ~13% typical scatter, built into the simulation as random measurement noise. The point of the table is what happens in aggregate: 23 independent, individually noisy estimates average out much closer to the true k_B than any one row does on its own. This is the same logic Johnson relied on to report a credible value of Boltzmann’s constant from instrumentation that could not pin it down precisely in any single trial.

Table 1: Synthetic analogue of Johnson’s Table I: Boltzmann-constant determinations

	run	f0_Hz	T_K	R0_ohm	k_B_fit	pct_error
0	1	1170.0968	301	229743.6514	0.0	11.7696
1	2	1019.6550	301	468279.2227	0.0	4.7394
2	3	346.8505	301	584328.9819	0.0	-2.1178
3	4	815.4312	297	220637.5275	0.0	-3.8019
4	5	645.8739	295	775328.2054	0.0	16.8228
5	6	1967.2532	297	752310.9467	0.0	-2.2720
6	7	770.7150	298	972932.8719	0.0	27.5319
7	8	1359.9326	298	651702.9709	0.0	8.4071
8	9	367.3079	297	513402.2946	0.0	1.4172
9	10	1749.4758	295	334087.7030	0.0	-12.2818
10	11	1166.1430	295	777727.1869	0.0	7.7187
11	12	1461.5877	295	272454.6331	0.0	9.5115
12	13	625.2507	300	869704.2769	0.0	-0.2608
13	14	1102.2465	301	106382.6457	0.0	3.5463
14	15	1523.8460	298	353690.0446	0.0	3.0616
15	16	1668.5932	297	235472.3474	0.0	15.9204
16	17	1821.0170	301	630551.8559	0.0	11.6465
17	18	1862.4507	295	896968.2400	0.0	-5.7958
18	19	717.4389	301	290507.2683	0.0	11.4893
19	20	1703.3293	301	437632.2968	0.0	5.7925
20	21	1475.2730	298	456630.5460	0.0	-3.3650
21	22	1016.0210	295	669843.9514	0.0	0.4290

Table 1: Synthetic analogue of Johnson’s Table I: Boltzmann-constant determinations

	run	f0_Hz	T_K	R0_ohm	k_B_fit	pct_error
22	23	1532.9997	301	488104.0739	0.0	-1.6327

mean k_B = 1.4456e-23 J/K (true = 1.3806e-23)
mean |err| = 7.4%

Johnson reported a mean deviation of about 13% across his 23 determinations; the synthetic run above was constructed with that same scatter, so recovering a comparable mean error here is closer to a consistency check on the simulation than an independent confirmation.

6 Nyquist’s thermodynamic argument

Nyquist derives the same law with no reference to amplifiers or measurement: from the transmission-line construction, two resistors R at temperature T , joined by a non-dissipative line matched to R ($\sqrt{L/C} = R$), exchange power via traveling waves.

Shorting both ends at some instant traps the line’s energy, which is then equivalently described as a discrete set of standing-wave normal modes of a line of length l with propagation velocity v : mode frequencies $nv/2l$, so the number of modes in a band $d\nu$ is $2l d\nu/v$. By equipartition (see the background section above), each mode – being a harmonic oscillator, exactly the case equipartition covers – carries a full $k_B T$ on average, split evenly between its electric and magnetic energy. This gives a trapped energy $2lk_B T d\nu/v$ in that band – accumulated over the transit time l/v – hence available power $k_B T d\nu$ per conductor. The length l and velocity v are calculation scaffolding, not physical parameters of the final result: they enter both the mode-counting step and the transit-time normalization, and must cancel exactly for the argument to make sense (checked numerically below).

Combined with the circuit relation between EMF and dissipated power in the matched loop, this gives Eq. (1) of Nyquist’s paper, the same flat spectral density stated in the introduction:

$$\overline{E_\nu^2} = 4Rk_B T.$$

modes per Hz ($2l/v$)	= 1.000e-08
energy in 1 Hz band	= 4.142e-29 J
available power = energy / transit_time	= 8.284e-21 W/Hz
k_B * dnu	= 1.381e-23 W/Hz

The available power matches $k_B d\nu$ regardless of the arbitrary choices of l and v above, confirming that both cancel out of the final result as the derivation requires.

7 Detailed balance / the second-law argument

Before fixing the functional form, Nyquist first argues the noise EMF must be a *universal* function of ν , R , T only – not merely on average, but in every frequency band separately.

The argument works by contradiction: assume the opposite of what is being claimed, and show that the assumption leads to something physically impossible – which means the assumption must be false. Concretely: suppose two conductors at the same temperature exchanged *more* power in frequency band A from conductor I to II than the reverse (i.e., suppose the noise power is *not* balanced in that band, even though it is balanced in total). Insert a non-dissipative resonant filter tuned to pass only band A between the two conductors. If that imbalance in band A were real, the filter would now let a net power flow from II to I (or I to II) – even though the two conductors are at the same temperature and nothing is driving that flow. That amounts to extracting usable work from a system sitting in thermal equilibrium with no temperature difference to drive it, which the second law of thermodynamics rules out. Since the assumed imbalance leads to something impossible, no such imbalance can exist: the noise power must be balanced in every frequency band separately, not just on average across all of them.

This is a purely conceptual argument (no computation needed) but it is the step that licenses treating E_ν^2 as a function of ν alone, rather than only $\int E_\nu^2 d\nu$ in total – the detailed-balance idea reappears throughout statistical mechanics (see `detailed_balance.pdf` in this folder).

8 Generalization to arbitrary networks

Nyquist extends the flat-resistor result to any passive network with complex impedance $R_\nu + iX_\nu$ at temperature T :

$$\overline{E_\nu^2} = 4R_\nu k_B T \quad (\text{Nyquist's Eq. 4})$$

Here R_ν is genuinely frequency dependent for a general network, so $\overline{E_\nu^2}$ is no longer flat – this is the one place before the quantum section where a ν -dependent quantity actually appears on the right-hand side. Reduced through the transfer admittance $Y(\omega)$ of a measuring network, this becomes

$$\overline{I^2} = \frac{2}{\pi} k_B T \int_0^\infty R(\omega) |Y(\omega)|^2 d\omega \quad (\text{Nyquist's Eq. 6} = \text{Johnson's Eq. 1}),$$

exactly the integral used in the Boltzmann-constant section above – the two papers’ central equations are the same formula, arrived at from opposite directions.

This is also the point where the fluctuation-dissipation theorem is on full display in its general form: R_ν , a dissipative response (how strongly the network absorbs and converts energy to heat at frequency ν), fixes $\overline{E_\nu^2}$, an equilibrium fluctuation (the spontaneous voltage noise generated at that same frequency) – for *any* passive network, not just a bare resistor. Nothing about the network’s internal construction matters beyond the single number R_ν it presents at each frequency.

9 Classical-to-quantum crossover

Nyquist notes, without pursuing it far, that replacing the equipartition energy $k_B T$ per mode with the Planck oscillator average

$$\langle \epsilon \rangle = \frac{h\nu}{e^{h\nu/k_B T} - 1}$$

gives the exact spectral density

$$\overline{E_\nu^2} = \frac{4R_\nu h\nu}{e^{h\nu/k_B T} - 1} \quad (\text{Nyquist's Eq. 8}),$$

reducing to the classical result for $h\nu \ll k_B T$. In 1928 this was experimentally indistinguishable from the classical law at any accessible frequency and temperature; it becomes relevant at optical/infrared frequencies and low temperature, which is presumably why it’s worth carrying into a detector-noise context. The crossover falls at $\nu = k_B T/h$, which at room temperature sits in the far infrared.

`crossover frequency  $k_B T / h = 6.251\text{e}+12$  Hz    (T = 300 K)`

Repeating the quantum spectrum at cryogenic temperatures shows the crossover frequency scaling linearly with T , pushing the rolloff to lower frequency as the system is cooled.

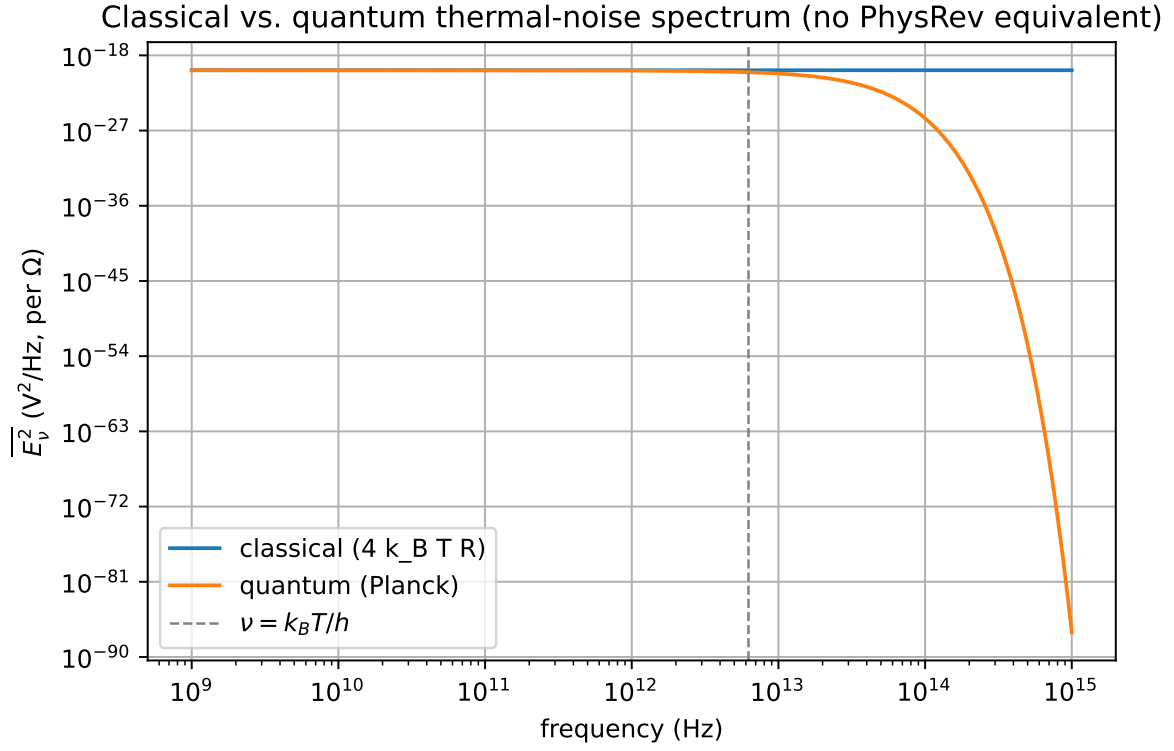


Figure 6: Classical vs. quantum thermal-noise spectrum, $T = 300$ K (no corresponding PhysRev figure – illustrates Nyquist's Eq. 8)

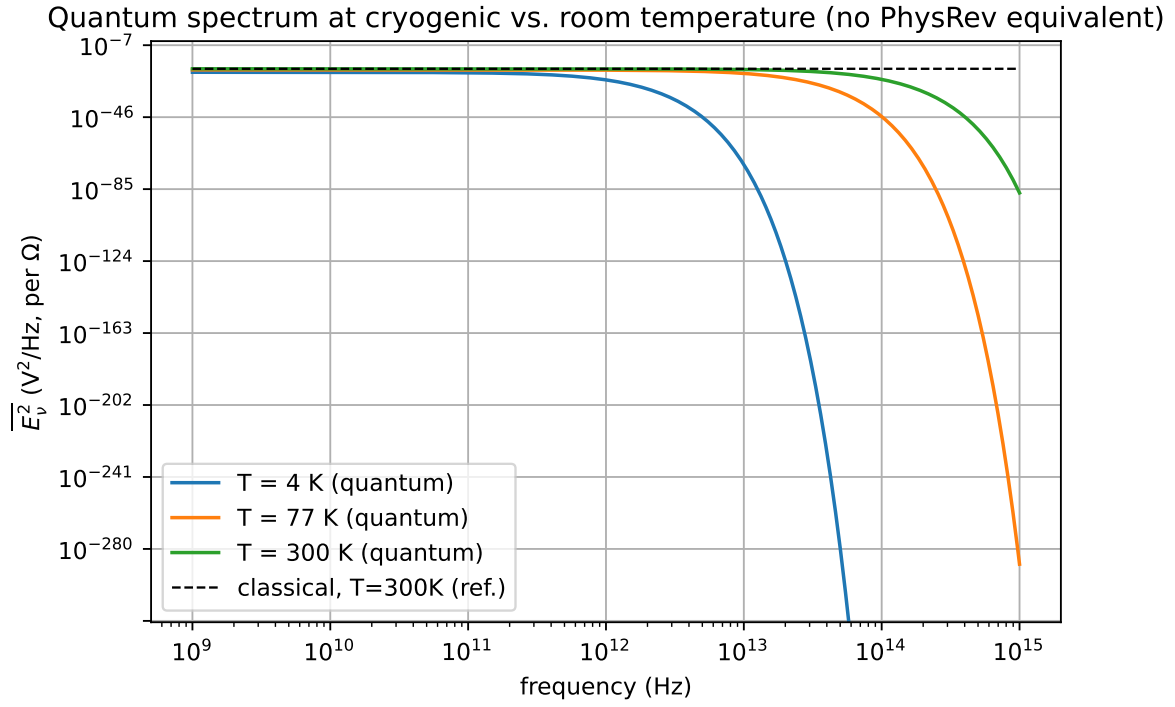


Figure 7: Quantum spectrum at cryogenic vs. room temperature (no corresponding PhysRev figure – illustrates Nyquist’s Eq. 8)

10 Synthesis and further reading

Johnson isolated a resistance- and temperature-dependent noise floor empirically, using comparative measurements across deliberately heterogeneous conductors (metallic films, electrolytes, carbon) to eliminate every variable but R and T , and used it to extract k_B . Nyquist derived the same spectral law from two ingredients alone – equipartition (Section 2 background; invoked directly in Section 5) and detailed balance (Section 6) – with no reference to the microscopic carriers at all.

This combination is not specific to electrical noise: it is the historical prototype of the **fluctuation-dissipation theorem**, the general statement that a system’s equilibrium fluctuations are fixed by its own dissipative response, with temperature as the only other ingredient. Wherever a system can lose energy to a thermal bath, that same bath must be driving it randomly, and the two effects are locked together by exactly one relation – here, $\overline{E_\nu^2} = 4R_\nu k_B T$. The same reasoning fixes the mean-square displacement of a Brownian particle in terms of its friction coefficient, and, closer to an imaging-science context, fixes the achievable noise floor of a detector or amplifier stage in terms of its own dissipative loss mechanisms – which is precisely why fluctuation-dissipation reasoning, not just this one electrical formula, underlies noise-equivalent-power and detectivity calculations generally.

Related papers:

- `fluctuation_dissipation.pdf` – the general theorem this result is a special case of.
- `detailed_balance.pdf` – the equilibrium argument used above, in general form.
- `equipartition_theorem.pdf` – background for the $k_B T$ -per-mode step above.
- `Noise_Art_Of_Electronics.pdf`, `More_noise.pdf`, `even_more_Noise.pdf`, `Detectors_Noise_montana.pdf` – applied/engineering treatments, useful for connecting this to detector noise-equivalent-power calculations.